

The functions $y = \arcsin x$, $y = \arccos x$, $y = \arctan x$, $y = \operatorname{arccot} x$

The four inverse curves — their shapes, symmetries and asymptotes, obtained by one reflection.

LESSON 75–76

2 × 40 MIN

ALGEBRA 10, §4.3

P1 · 5.3

LESSON CLOCK

12'

24'

22'

22'

16'

8'

One reflection, four curves	12 min
Arcsin and arccos	24 min
Arctan and arccot	22 min
Transformations	22 min
Practice	16 min
Homework	8 min

LEARNING OBJECTIVES

- Sketch each inverse trigonometric function from the restricted original.
- State the domain, range, monotonicity and symmetry of each.
- Identify the asymptotes of arctan and arccot.
- Apply transformations to the inverse curves.

TEXTBOOK

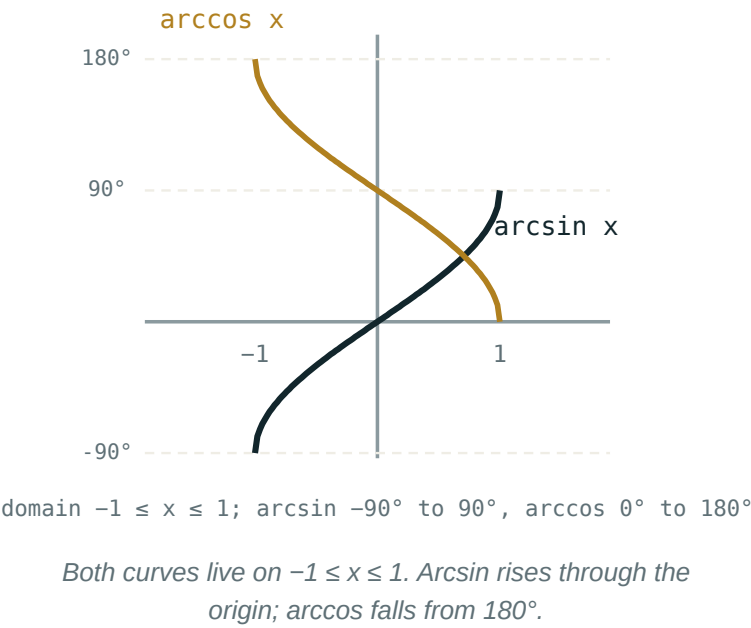
National	pp. 325–336
Cambridge	Pure Mathematics 1, pp. 105–110
Workbook	P1 Exercise 5C

01

Explanation

One reflection, four curves

Take the restricted branch of each trigonometric function and reflect it in $y = x$. The domain and range swap; increasing stays increasing; the x - and y -intercepts exchange.



EVERYTHING FOLLOWS FROM THE ORIGINAL

A vertical asymptote of the original becomes a **horizontal** asymptote of the inverse. A maximum becomes a right-hand endpoint. There is nothing new to learn — only to reflect.

Arcsin and arccos

	$Y = \text{ARCSIN } X$	$Y = \text{ARCCOS } X$
domain	$[-1, 1]$	$[-1, 1]$
range	$[-90^\circ, 90^\circ]$	$[0^\circ, 180^\circ]$
behaviour	increasing	decreasing
through	$(0, 0)$	$(0, 90^\circ)$
endpoints	$(-1, -90^\circ), (1, 90^\circ)$	$(-1, 180^\circ), (1, 0^\circ)$
symmetry	odd — half-turn about O	half-turn about $(0, 90^\circ)$

The relation $\arcsin x + \arccos x = 90^\circ$ says the two curves are reflections of each other in the horizontal line $y = 45^\circ$ — a fact visible at a glance in the figure.

Arctan and arccot

	$Y = \operatorname{ARCTAN} X$	$Y = \operatorname{ARCCOT} X$
domain	$\mathbb{R}$	$\mathbb{R}$
range	$(-90^\circ, 90^\circ)$	$(0^\circ, 180^\circ)$
behaviour	increasing	decreasing
through	$(0, 0)$	$(0, 90^\circ)$
asymptotes	$y = \pm 90^\circ$	$y = 0^\circ, y = 180^\circ$
symmetry	odd	half-turn about $(0, 90^\circ)$

THE ASYMPTOTES ARE HORIZONTAL, AND NEVER REACHED

$\arctan x$ climbs towards 90° for large x but never attains it — because $\tan 90^\circ$ does not exist. The range is an **open** interval, which is why the endpoints are written with round brackets.

Transformations

The rules of Quarter I apply unchanged:

CURVE	DOMAIN	RANGE
$y = \arcsin x$	$[-1, 1]$	$[-90^\circ, 90^\circ]$
$y = \arcsin(2x)$	$[-0.5, 0.5]$	$[-90^\circ, 90^\circ]$
$y = 2 \arcsin x$	$[-1, 1]$	$[-180^\circ, 180^\circ]$
$y = \arcsin(x - 1)$	$[0, 2]$	$[-90^\circ, 90^\circ]$
$y = \arctan x + 90^\circ$	$\mathbb{R}$	$(0^\circ, 180^\circ)$

The last row is worth noticing: $\arctan x + 90^\circ$ has exactly the range of $\operatorname{arccot} x$, and in fact $\operatorname{arccot} x = 90^\circ - \arctan x$ — the two are reflections in a horizontal line, as the arcsine and arccosine were.

02 Worked examples

EXAMPLE 1 State the domain and range of $y = \arccos(x - 2)$.

- ① Need $-1 \leq x - 2 \leq 1$.
- ② $1 \leq x \leq 3$
- ③ The range is unchanged by a horizontal shift.

ANSWER

Domain $[1, 3]$, range $[0^\circ, 180^\circ]$

EXAMPLE 2 State the domain and range of $y = 3 \arcsin(2x)$.

- ① $-1 \leq 2x \leq 1 \Rightarrow -0.5 \leq x \leq 0.5$
- ② The outer 3 stretches the range.
- ③ $3 \times [-90^\circ, 90^\circ]$

ANSWER

Domain $[-0.5, 0.5]$, range $[-270^\circ, 270^\circ]$

EXAMPLE 3 What are the asymptotes of $y = \arctan(x) - 30^\circ$?

① $\arctan x$ has asymptotes $y = \pm 90^\circ$.

② A vertical shift moves them by -30° .

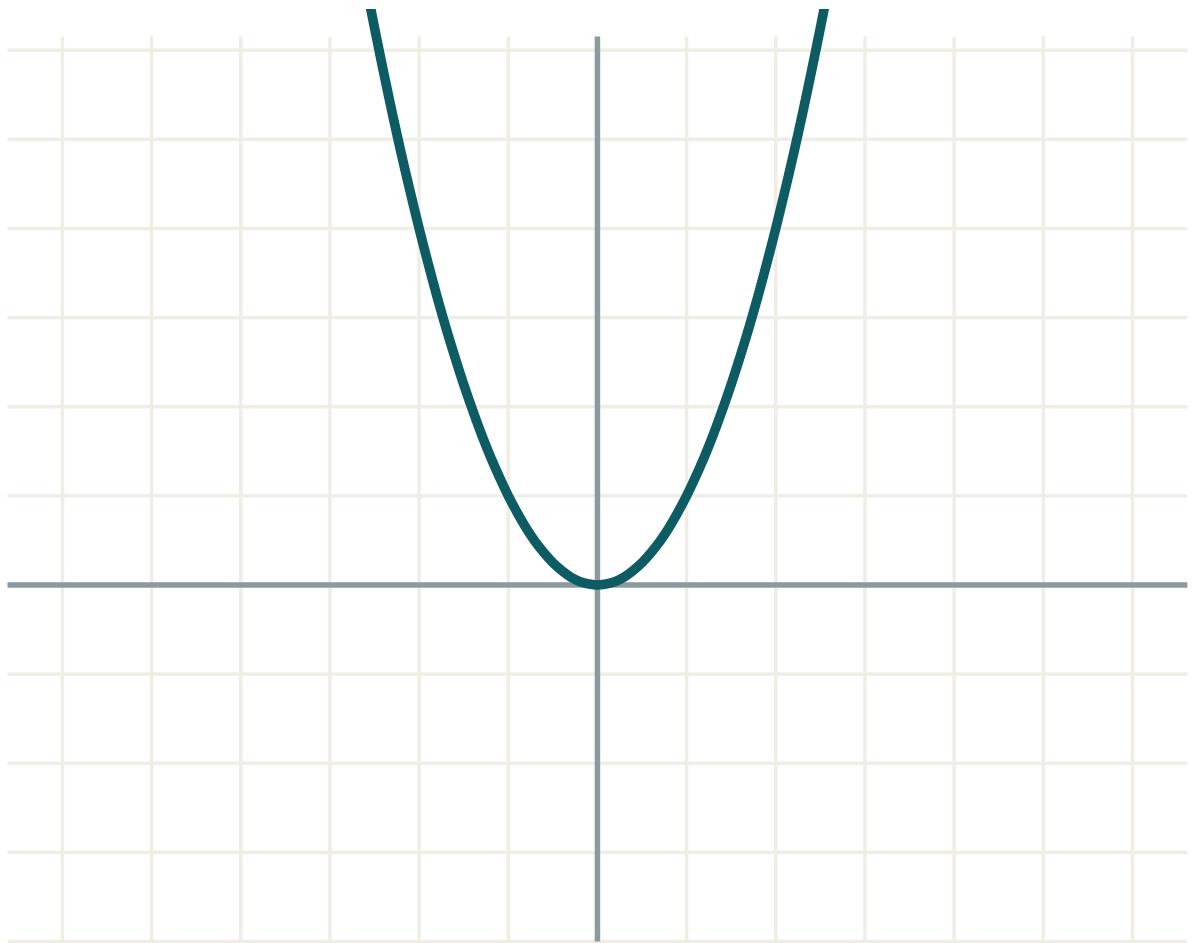
ANSWER

$$y = 60^\circ \text{ and } y = -120^\circ$$

03 **Interactive model**

Reflect the sine curve in $y = x$ with tracing paper — the arcsine appears exactly.

Reflecting a curve



a 1

b 0

c 0

equation $y = f(x)$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Inverse function	Teskari funksiya	Обратная функция
Reflection in $y = x$	$y = x$ ga nisbatan akslantirish	Симметрия относительно $y = x$
Restricted branch	Cheklangan tarmoq	Ограниченная ветвь
Horizontal asymptote	Gorizontal asimptota	Горизонтальная асимптота
Increasing function	O'suvchi funksiya	Возрастающая функция
Decreasing function	Kamayuvchi funksiya	Убывающая функция
Odd function	Toq funksiya	Нечётная функция
Centre of symmetry	Simmetriya markazi	Центр симметрии
Point of inflection	Egilish nuqtasi	Точка перегиба

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The domain of $\arcsin x$ is:

$\arccos x$ is:

The asymptotes of $\arctan x$ are:

The domain of $\arcsin(2x)$ is:

$[-1, 1]$

$[-0.5, 0.5]$

$[-2, 2]$

$\mathbb{R}$

The range of $2 \arccos x$ is:

$[0^\circ, 180^\circ]$

$[0^\circ, 360^\circ]$

$[-180^\circ, 180^\circ]$

$[0^\circ, 90^\circ]$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Domain of $\arcsin x$

ANSWER $[-1, 1]$

2. Range of $\arcsin x$

ANSWER $[-90^\circ, 90^\circ]$

3. Range of $\arccos x$

ANSWER $[0^\circ, 180^\circ]$

4. Range of $\arctan x$

ANSWER $(-90^\circ, 90^\circ)$

5. Is $\arcsin x$ increasing?

ANSWER yes

6. Is $\arccos x$ increasing?

ANSWER no

7. $\arccos x$ passes through

ANSWER $(0, 90^\circ)$

Medium · the standard question

7 PROBLEMS

1. Domain of $\arccos(x - 2)$

ANSWER $[1, 3]$

2. Domain of $\arcsin(2x)$

ANSWER $[-0.5, 0.5]$

3. Range of $3 \arcsin x$

ANSWER $[-270^\circ, 270^\circ]$

4. Range of $\arctan x + 90^\circ$

ANSWER $(0^\circ, 180^\circ)$

5. Asymptotes of $\arctan x - 30^\circ$

ANSWER $y = 60^\circ$ and $y = -120^\circ$

6. Domain of $\arcsin(x + 1)$

ANSWER $[-2, 0]$

7. Range of $-\arccos x$

ANSWER $[-180^\circ, 0^\circ]$

Hard · stretch and reason

7 PROBLEMS

1. Domain and range of $y = \arcsin(3x - 1)$

ANSWER $[0, \frac{2}{3}], [-90^\circ, 90^\circ]$

2. Domain and range of $y = 2 \arccos(\frac{x}{2}) - 90^\circ$

ANSWER $[-2, 2], [-90^\circ, 270^\circ]$

3. Show $\operatorname{arccot} x = 90^\circ - \arctan x$

ANSWER Complementary, with matching ranges

4. Sketch $y = \arcsin x$ and $y = \arccos x$ and mark the line of symmetry

ANSWER $y = 45^\circ$

5. Where do $\arcsin x$ and $\arccos x$ meet?

ANSWER $x = \frac{\sqrt{2}}{2}$, at 45°

6. Domain of $\arcsin(x^2)$

ANSWER $[-1, 1]$

7. Explain why the range of $\arctan$ is open at both ends

ANSWER $\tan(\pm 90^\circ)$ does not exist

07 Homework

Homework — 5 tasks

Every sketch needs the endpoints or the asymptotes marked.

- Sketch $y = \arcsin x$ and $y = \arccos x$ on one set of axes, with endpoints.
- Sketch $y = \arctan x$ with its asymptotes.
- State the domain and range of $y = \arccos(2x + 1)$ and $y = 2 \arcsin(x - 1)$.
- Find where $y = \arcsin x$ and $y = \arccos x$ intersect.

5. Show that $\operatorname{arccot} x = 90^\circ - \operatorname{arctan} x$ and check both ranges agree.